# BMC Helix Technical Bootcamp - ITSM/ITOM Container

## Lab 2: Deploying ITOM

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# Lab 2: Deploying ITOM

**OBJECTIVE**: In this lab, you will use the Health Check Tool (HCT) and other methods to validate a customer's environment for installation and then use the Helix installer to deploy one or more applications and troubleshoot installation issues.

Refer to the full product documentation for complete implementation details. The steps provided in this lab document are simplified and tailored to our specific environment. It is recommended that you review the full documentation before beginning any lab work: https://docs.bmc.com/xwiki/bin/view/Service-Management/On-Premises-Deployment/BMC-Helix-Service-Management-Deployment/brid25101/Installing/Preparing-for-installation/Installing-BMC-Helix-Platform-Common-services-25-1-00/

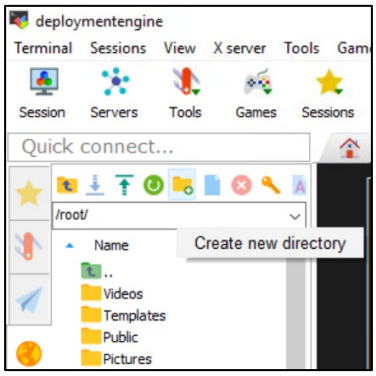
## Task 1: Copy the Installation Files to the Deployment Engine

In a customer implementation, installation files would be downloaded from EPD. In this task, the installation files have been provided for you on the system but need to be copied to the appropriate directory on the Deployment Engine.

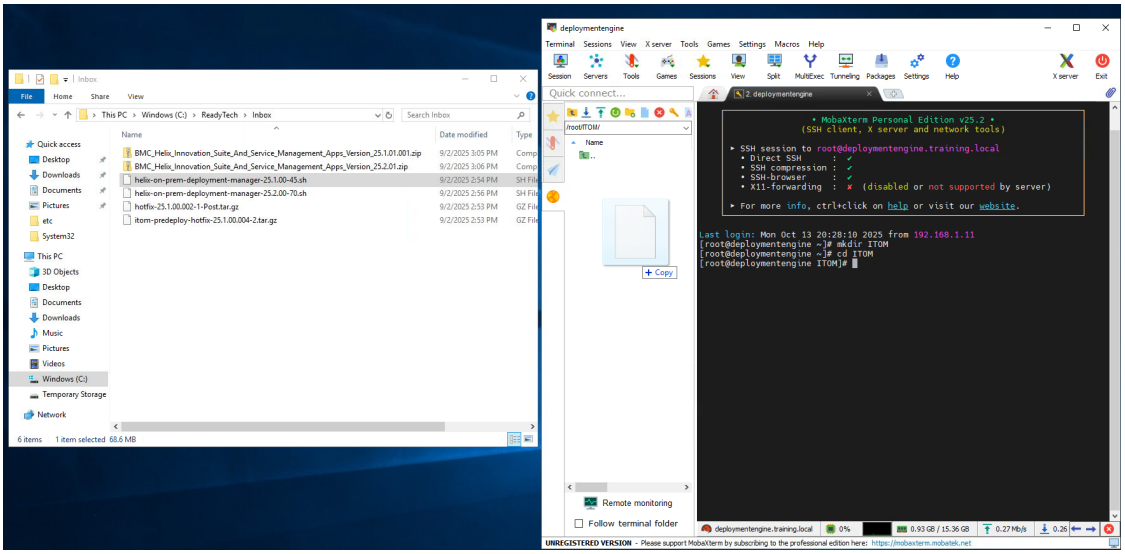
1. Create a directory on **deploymentengine** and copy the helix-on-prem-deployment-manager-25.1.00-45.sh file from **C:\ReadyTech\Inbox** into the directory. You can do that either in MobaXterm or WinSCP.

   For **MobaXterm**:

   - Make sure you are connected to **deploymentengine** and log in with credentials **root/password**.
   - Create a directory called **ITOM** in the **/root** directory using the **Create new directory** button in the SSH Browser (SCP) panel.



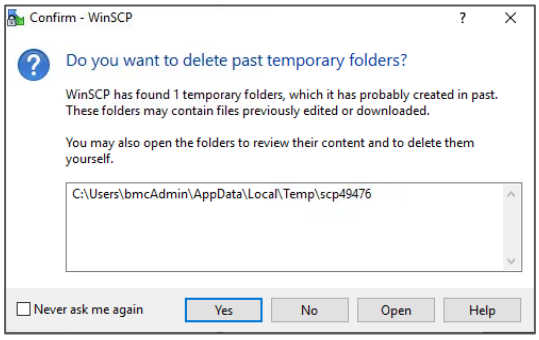

   - Go to the **C:\ReadyTech\Inbox** folder on the landing desktop and drag **helix-on-prem-deployment-manager-25.1.00-45.sh** to **/root/ITOM** folder. You may need to refresh your MobaXterm folder view.
   - At the bottom left of the MobaXterm screen you can monitor the progress of the copy.



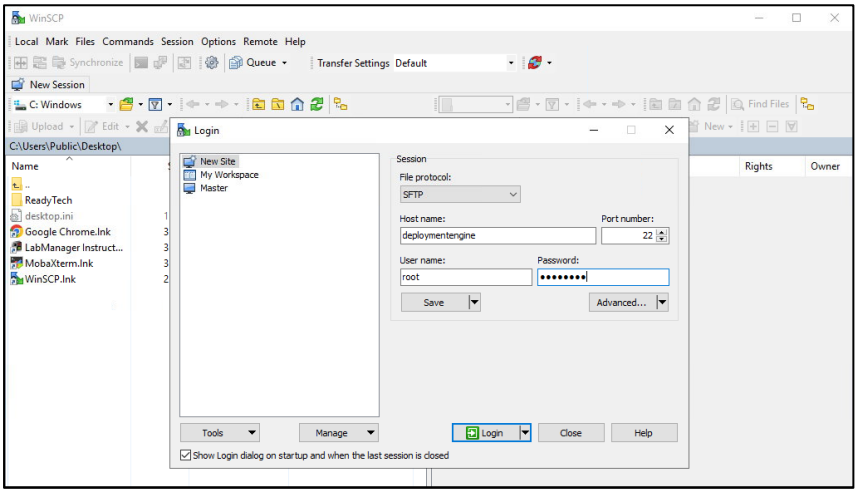

Or if you prefer **WinSCP**:

- Open WinSCP from the Windows Desktop

- Click **Yes** when prompted to delete temporary folders. This is a left over artifact on our system that is unrelated to the installation of the ITOM product.

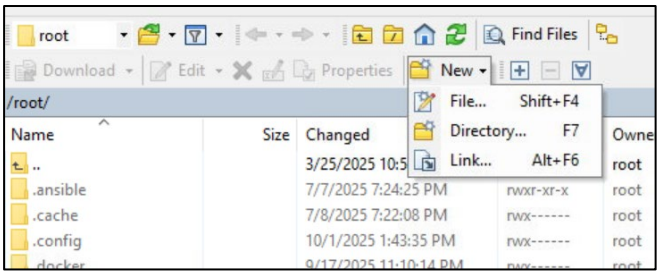



- Click on **New Site** and enter deploymentengine in Host name.
- Username: **root** /Password: **password**
- Click **Login**.




- Click **Yes** on the Warning pop-up.

- Create a new directory called **ITOM** in root.




- Drag **C:\ReadyTech\Inbox\helix-on-prem-deployment-manager-25.1.00-45.sh** to **/root/ITOM** folder.

## Task 2: Extract the Installer

Before you can install the Helix Platform Common Services, you must unpack the installer files from the self-extracting archive and accept the license agreement.

1. Go to the **/root/ITOM** directory and set the correct permissions on the helix-on-prem-deployment-manager-25.1.00-45.sh file with this command:

   **`chmod a+x helix-on-prem-deployment-manager-25.1.00-45.sh`**

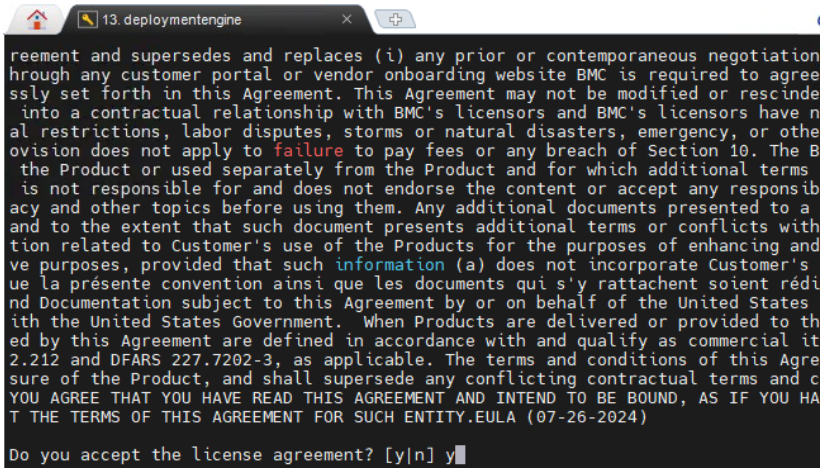
   ```
   [root@deploymentengine ~]# cd /root/ITOM
   [root@deploymentengine ITOM]# chmod a+x helix-on-prem-deployment-manager-25.1.00-45.sh
   [root@deploymentengine ITOM]#
   ```

   **Note:** This command will give all users permissions to execute the specified file.

2. Unpack the installer files from the "helix-on-prem-deployment-manager-25.1.00-45.sh" self-extracting archive by running this command:

   **`./helix-on-prem-deployment-manager-25.1.00-45.sh`**

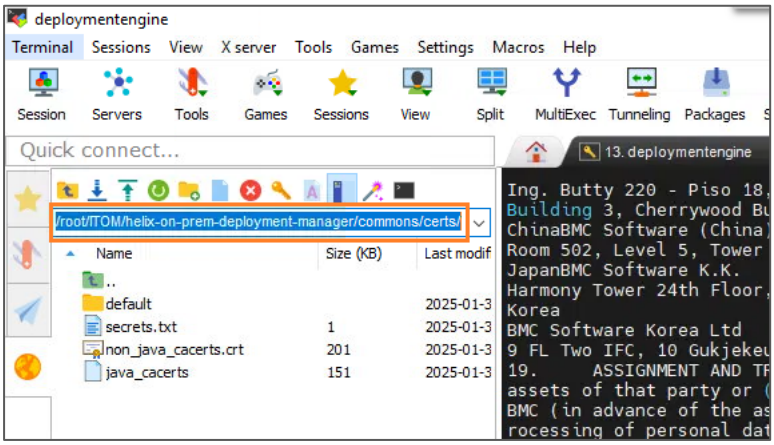
3. Type **y** and then **Enter** to accept the **License Agreement**.



## Task 3: Update secrets.txt file

The secrets.txt file must be updated with the appropriate values to prepare for password encryption.

1. Using **MobaXTerm**, go to the **/root/ITOM/helix-on-prem-deployment-manager/commons/certs** folder.

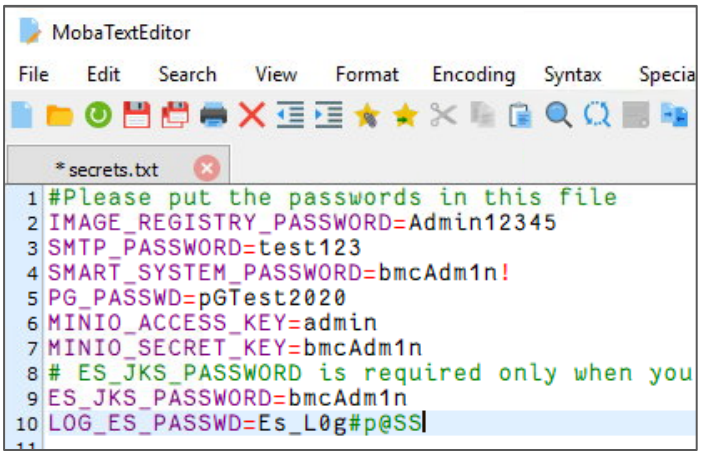



2. Locate the **secrets.txt** file. This file is used to generate an encrypted secret.config file. This configuration file allows for storage of key secrets (vs. in the infra.config, etc.), allowing for better separation during installation process and restricting the need for plaintext passwords. The secrets.txt file is required by the Health Check Tool (HCT).

3. Double-click the **secrets.txt** file to edit with the values below using **MobaTextEditor**. Do not edit or remove the commented lines as they are used by the installer and the HCT.
   - IMAGE_REGISTRY_PASSWORD=**Admin12345**
   - SMTP_PASSWORD=**test123**
   - SMART_SYSTEM_PASSWORD=**bmcAdm1n!**
   - PG_PASSWD=**pGTest2020**
   - MINIO_ACCESS_KEY=**admin**
   - MINIO_SECRET_KEY=**bmcAdm1n**
   - ES_JKS_PASSWORD=**bmcAdm1n**
   - LOG_ES_PASSWD=**Es_L0g#p@SS**

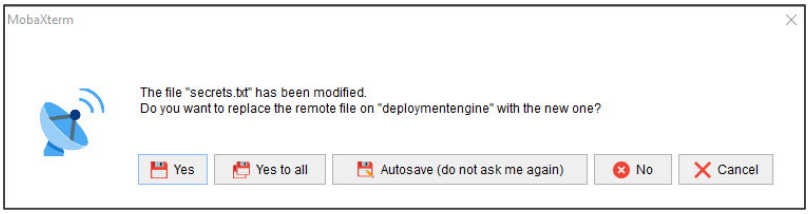
   **Note:** For SMART_SYSTEM_PASSWORD, ensure to use the password indicated above or the installation will fail, this value is used for the connection to BMC Discovery.

4. Verify your **secrets.txt** file matches the screenshot below. Remove any quotation marks that are in the file.




5. **Save** the file.

6. Click Yes to confirm with MobaXterm to replace the remote file on the Deployment Engine.




7. Go to the directory that contains the secrets.txt file and create a backup of the file by running this command:

   ```
   cp secrets.txt  secrets.bak
   ```

   **Note:** It's a best practice to make a backup copy of the secrets.txt file after editing because the secrets.txt file is encrypted and then deleted after installation.

## Task 4: Update infra.config file

The infra.config file contains more than forty items that need to be configured correctly to install the Helix Platform and related solutions. We expect these to continue to streamline over time.

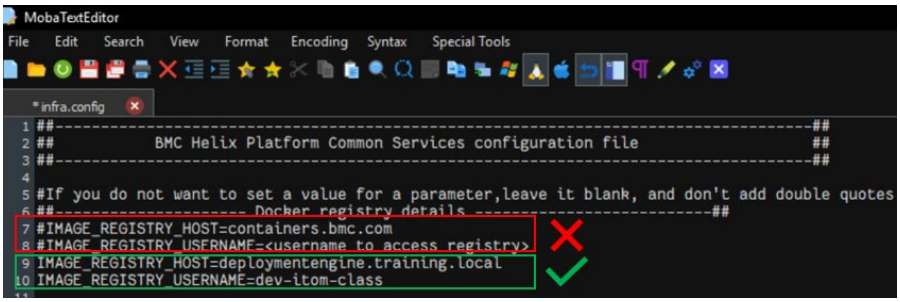
1. Using **MobaXTerm SSH browser (SCP)** tab, go to the **/root/ITOM/helix-on-prem-deployment-manager/configs** folder.

2. Locate the **infra.config** file.

3. Double-click the **infra.config** file to edit, using **MobaTextEditor**. Below is an annotated list to help you configure the file. The parameters are not listed in the same order as in the file; instead, they're grouped by common topics (such as hostnames) to make them easier to understand.

   When editing the infra.config file keep these things in mind:

- Use **MobaTextEditor** or the **VIM** editor as Windows editors can cause corruption.
- Unless specifically noted, all values should be in lowercase.
- Make sure there are no spaces, for example: IMAGE_REGISTRY_USERNAME= dev-itom-class
- Make sure you **do not edit the comment # lines** in the file. You need to edit the lines without a # comment.




Here are the first set of parameters that need to be configured in the infra.config file:

| Parameter | Value | Notes |
|---|---|---|
| IMAGE_REGISTRY_HOST | deploymentengine.training.local | Image Registry Host |
| IMAGE_REGISTRY_USERNAME | dev-itom-class | Image Registry User Name |
| NAMESPACE | helixade | This is the namespace used for base product installation. |
| LB_HOST | adelab.training.local | This is the main URL used to access the solution. |
| LB_PORT | 443 | Keep this value for the lab. This is pre-provisioned |
| TMS_LB_HOST | tms.training.local | |
| DOMAIN | training.local | This is the subdomain used for all other hostnames within the solution. |
| MINIO_LB_HOST | minio.training.local | |
| MINIO_API_LB_HOST | minio-api.training.local | |
| TENANT_ENVIRONMENT | poc | |
| CLUSTER_TYPE | | Do not set |
| TENANT_NAME | adelab | Lowercase |
| TENANT_EMAIL | joe.bloggs@training.local | |
| TENANT_FIRST_NAME | Joe | Case does not matter |
| TENANT_LAST_NAME | Bloggs | Case does not matter |
| TENANT_TYPE | private | Use lowercase letters |
| TENANT_COUNTRY | "United States" | Must fit a specific list of countries, from here. Must be enclosed in double quotes, and case must match the list. |

4. To successfully send an email, the relevant SMTP parameters need to be updated in the environment. Once the product install is successful, the installer sends an email to the

customer/tenant with login credentials and URL. For testing purposes, we are using a MailHog web client on the Windows server.

Here are the **SMTP** parameters that need to be configured in the infra.config file:

| Parameter | Value | Notes |
|---|---|---|
| **SMTP_HOST** | 192.168.1.20 | This value will send all email to the Mailhog service. |
| **SMTP_PORT** | 25 | This is the tcp network port used for the SMTP_HOST. 25 is typical for internal, unencrypted mail servers, while there may be other ports provided by the customer when using encryption. |
| **SMTP_USERNAME** | "" | Use double quotes. |
| **SMTP_FROM_EMAIL** | helix-activation@example.com | This is the "From:" address on the notification email. The customer will usually have a team email or contact they want to use. |
| **SMTP_TLS** | false | The SMTP server TLS, we will not be setting this. |
| **SMTP_AUTH_DASHBOARD** | true | This variable is used for getting report email to dashboard team by default value is true<br>SMTP_AUTH_DASHBOARD=<true or false> |
| **SMTP_AUTH** | NONE | This value is case sensitive. Use this value when you want to skip SMTP authentication. If you set the value as NONE, set the username as shown below:<br>SMTP_USERNAME=""<br><br>**NOTE:** Configuring SMTP_AUTH with just "" will result in an error in this version, see this KA:<br>https://community.bmc.com/s/article/BMC-Helix-Portal-OnPrem-An-error-occurred-Please-contact-your-administrator-or-retry-later-when-logging-in-after-setting-hannah-admin-password |
| **OPS_GROUP_EMAIL** | ops@training.local | Use an email address you can receive email at. Since we're using Mailhog in this lab, it doesn't matter, but when doing an implementation, all of these addresses should be valid. |
| **APPROVAL_GROUP_EMAIL** | approval@training.local | Use an email address you can receive email at. Since we're using Mailhog in this lab, it doesn't matter but when doing an implementation, all of these addresses should be valid. |

5. Block storage is recommended; however, we are using NFS for simplicity within this lab. A NFS provisioner is deployed in the lab to present the NFS volume to Kubernetes.

   A few items to note about these parameters:

   - BMC supports a Bring-Your-Own-Storage-Class model, for any block storage supporting high performance IOPS. NFS is NOT supported in production for persistent volumes except for the storage classes that require read-write-many storage. CephRBD is certified by BMC.

   Here are the **Storage Provider** variables that need to be configured in the infra.config file:

| Parameter | Value | Notes |
|---|---|---|
| **PG_STORAGE_CLASS** | nfs-external | Most of these storage classes would normally be a block storage class but can be configured individually for different services. |
| **VMSTORAGE_STORAGE_CLASS** | nfs-external | |
| **VMAGGSTORAGE_STORAGE_CLASS** | nfs-external | |
| **ES_MASTER_STORAGE_CLASS** | nfs-external | |
| **ES_DATA_STORAGE_CLASS** | nfs-external | |
| **MINIO_STORAGE_CLASS** | nfs-external | |
| **EFS_STORAGE_CLASS** | nfs-external | |
| **REDIS_HA_GLOBAL_STORAGECLASS** | nfs-external | |
| **KAFKA_STORAGECLASS** | nfs-external | |
| **AIOPS_STORAGE_CLASS** | nfs-external | |
| **REDIS_CLUSTER_STORAGE_CLASS** | nfs-external | |
| **OPT_STORAGE_CLASS** | nfs-external | This storage class is used by the Capacity (BHCO) product. |

6. There are several Kubernetes specific configurations (and a couple that end up in this section since they have nowhere else to go, like the Smart System (Discovery) configuration)

   A few items to note about these parameters:

   - The NAMESPACE is where all the Helix Common Services (AKA PCS) components will be installed. We will be using "helixade", but in a customer environment, you may have a namespace assigned to you. We have Cluster Admin permission in the labs however we also support a custom service account/role/role binding in situations where Cluster Admin is not allowed.
   - CUSTOM_CA_SIGNED_CERT_IN_USE is set to true when using custom certificates. It is set to false when using a commercially purchased certificate. Most lab and POC users will be using a custom certificate and will set this to true.
   - CLUSTER_TYPE is used for openshift or ocp clusters. It is left blank for basic Kubernetes clusters.

Here are the **Kubernetes** variables that need to be configured in the infra.config file:

| Parameter | Value | Notes |
|---|---|---|
| **CUSTOM_CA_SIGNED_CERT_IN_USE** | true | We will need to provide a custom_cacert.pem file into the commons/certs/ folder to provide this certificate. |
| **CUSTOM_SERVICEACCOUNT_NAME** | helix-onprem-sa | Service account used to deploy Fluentbit. |
| **RSSO_CUSTOM_JAVA_KEYSTORE_IN_USE** | false | Custom JAVA keystore for "RSSO SAML keystore configuration" |
| **SMART_SYSTEM_USERNAME** | system | Smart Graph (Discovery) |
| **INGRESS_CLASS** | nginx | Used by the installer to create Ingress objects for application mapping. If set wrong, RSSO will break. |
| **NGINX_CONTROLLER** | NGINX_OPENSOURCE | |
| **INGRESS_TLS_SECRET_NAME** | "" | Two double quotes |
| **HELM_BIN** | /usr/local/bin/helm | Path to “helm” command line utility. |
| **KUBECTL_BIN** | /usr/local/bin/kubectl | Path to Kubernetes command line utility. |
| **OC_BIN** | | Path to openshift/ocp “oc” command line utility. Leave as blank |
| **RUN_AS_USER** | null | |
| **RUN_AS_GROUP** | null | |
| **FS_GROUP** | null | |
| **OPT_FSGROUP** | 1001 | |
| **ML_FSGROUP** | 998 | |

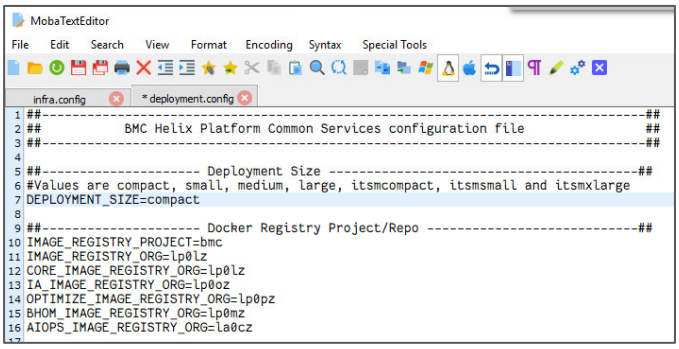
7. Save the **infra.config** file.

## Task 5: Update deployment.config file

The deployment.config file helps identify where to acquire container images, what size the install will be, and which repos/projects are to be used for which application components. In addition, the deployment.config file allows us to select which product components to install.

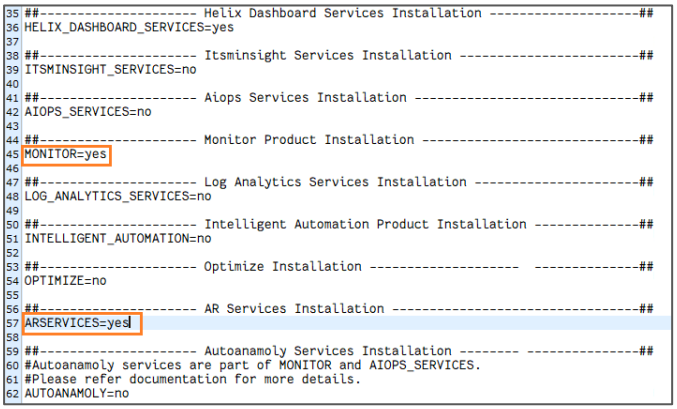
1. Go to **/root/ITOM/helix-on-prem-deployment-manager/configs** folder.

2. Edit the **deployment.config** file to set the **DEPLOYMENT_SIZE** to **compact**. Changing the DEPLOYMENT_SIZE to compact will reflect the combined ITSM/BHOM small environment we are deploying in. This will use the minimum resources for our training environment. This sizing is generally not used outside of labs or testing environments.




**NOTE:** In the Docker Registry Project/Repo section is where we may configure the image repository and project names if they are different from those set on containers.bmc.com. For the purposes of this class, they are the same as containers.bmc.com: top level project "bmc", repository "lp0lz", "lp0mz", etc. However, in some customer environments, the "bmc" top level label and even the specific repository names could be different.

3. To install the necessary product components, we need to set **MONITOR** and **ARSERVICES** to **yes** in **deployment.config** file. Setting MONITOR=yes, will ensure that BHOM is installed in addition to Common Services. Settings ARSERVICES=yes, will register the BMC Helix Innovation Suite services in BMC Helix Platform.




**NOTE:** We will not be installing ITSM Insights, AIOPS, Intelligent Automation, Optimize or Helix logging.

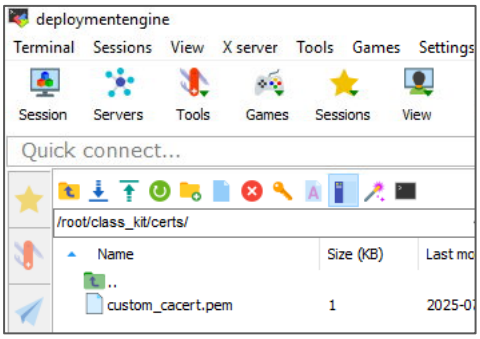
4. Save the **deployment.config** file.

## Task 6: Generate Custom (self-signed) certificate

For purpose of this installation, we are using a custom (self-signed) certificate that has already been generated. Production installs should generally be using CA-signed certificates provided by the customer.

1. Go to the **/root/class_kit/certs** directory.

2. Verify the **custom.cacert.pem** file is visible.




**NOTE:** The custom_cacerts.pem file should contain the signed certificate and the complete chain of CA certs that signed the certificate, e.g., signing, intermediate, and root. Since we are using a self-signed certificate (no CA) we can just copy the provided certificate to the commons/certs/custom_cacert.pem file.

3. Copy the self-signed certificate **custom_cacerts.pem** to **root/ITOM/helix-on-prem-deployment-manager/commons/certs** directory by using this command:

   ```
   cp /root/class_kit/certs/custom_cacert.pem /root/ITOM/helix-on-prem-deployment-manager/commons/certs/custom_cacert.pem
   ```

## Task 7: Run Health Check Tool (HCT)

Use the Health Check Tool (HCT) to verify your environment meets the configuration and tuning recommendations. By running this tool at different stages of your deployment, you can identify and fix issues.

1. Go to the **Health Check Tool** directory: **/root/class_kit/hct**

2. Edit the **sample-env-setting.sh** file. This file is used to set some environment variables that are used by the Health Check Tool (HCT).
   - INSTALL_DIR=**/root/ITOM/helix-on-prem-deployment-manager**
   - JAVA_HOME=**/usr/lib/jvm/jre-21-openjdk**
   - HCT_HOME=**/root/class_kit/hct**

3. The HCT will prompt for an IMAGE_REGISTRY_PASSWORD every time you run it. To pass it in automatically, add these lines to your **sample-env-setting.sh** file:
   - **IMAGE_REGISTRY_PASSWORD=Admin12345**
   - **export IMAGE_REGISTRY_PASSWORD**
   - **env| grep IMAGE_REGISTRY_PASSWORD**

```
#!/bin/bash
#INSTALL_DIR which is parent of configs directory
INSTALL_DIR=/root/ITOM/helix-on-prem-deployment-manager
export INSTALL_DIR

#Parent of bin directory of JRE
JAVA_HOME=/usr/lib/jvm/jre-21-openjdk
export JAVA_HOME

#HCT_HOME should be the extracted directory
#
HCT_HOME=/root/class_kit/hct
export HCT_HOME

SLEEP_DURATION_SECONDS=5
export SLEEP_DURATION_SECONDS

env|grep INSTALL_DIR
env|grep JAVA_HOME
env|grep HCT_HOME

IMAGE_REGISTRY_PASSWORD=Admin12345
export IMAGE_REGISTRY_PASSWORD
env| grep IMAGE_REGISTRY_PASSWORD
```

4. Save the **sample-env-setting.sh** file.

5. In the terminal window, go to the hct directory: **/root/class_kit/hct**

6. Source the sample-env-setting.sh file to set the environment variables so that you can run the HCT by using the following command:

   **`source sample-env-setting.sh`**

```
[root@deploymentengine hct]# source sample-env-setting.sh
INSTALL_DIR=/root/ITOM/helix-on-prem-deployment-manager
JAVA_HOME=/usr/lib/jvm/jre-21-openjdk
HCT_HOME=/root/class_kit/hct
IMAGE_REGISTRY_PASSWORD=Admin12345
[root@deploymentengine hct]#
```

   **Note:** This command will execute the sample-env-setting.sh script within the current shell session, applying any environment variable definitions, path updates, or configuration settings it contains so that they take effect immediately without starting a new shell.

7. Run the HCT in preinstall mode using this command:

   **`./helix-healthcheck.sh preinstall`**

8. Verify that after a long series of output the following message is displayed:

```
##########################################################################################
[251002-155055] INFO: Conclusion:- The system is ready for installation after reviewing the warning.
##########################################################################################
```

   **NOTE:** The following warnings/errors can be ignored:

   - *error: Required resource not specified*
   - *WARNING: Insufficient CPU count 87, required CPU 150. Chosen configuration is COMPACT. If you have defined Cluster Autoscaler, this message can be ignored.*
   - *WARNING: Insufficient memory 260786176 / 248 GB, required memory 327798360 / 312 GB. Chosen configuration is COMPACT.*

- *WARNING: HELM_VERSION=""3.16" "*

**Note:** Occasionally the tool may have difficulty pulling down the container for the nslookup pod which results in this error:

- *ERROR: validate-nslookup-daemonset pod=infra-client-daemonset-validate-nslookup-daemonset-rsmrx node=worker1 <none> ContainerCreating. Please refer <extract_dir>/hct/hct/logs/hct-deployment.log for more details*

To resolve, kick off the HCT tool again: ./helix-healthcheck.sh preinstall

9. If you have any other errors when you run HCT, you need to troubleshoot before continuing.

## Task 8: Install BMC Helix Platform Common Services

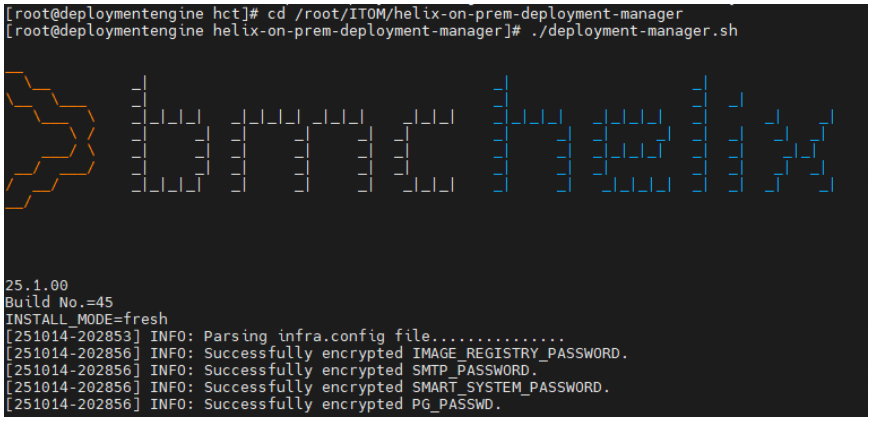
The Helix Platform installer as of the 25.x era consists of several shell scripts wrapping some pre-install checks (most of which have been satisfied by passing the Health Check Tool), a series of helm charts to build out the different k8s objects (statefulsets, deployments, services, etc.), and a set of jobs to configure the applications once successfully installed.

The installation will deploy the Helix Platform Core Services (configured as INFRA=yes in the deployment.config) to support other applications like Innovation Suite/ITSM, Monitor, Dashboards, Capacity, etc.

1. Start the BMC Helix Common Platform installer by running these two commands:

   **`cd /root/ITOM/helix-on-prem-deployment-manager`**

   **`./deployment-manager.sh`**




   **NOTE:** The install will run for ~2 hours. You should not need to interact with the installer again until it completes successfully. However, there are some things you may want to do while it is running, including monitoring the install for progress, timeliness, and any non-fatal but fixable errors.

2. To monitor progress of the product installation:

- You can look at the objects being created in the helixade namespace by running:

```
kubectl -n helixade get all
```

```
[root@deploymentengine ~]# kubectl -n helixade get all
NAME                                                      READY   STATUS      RESTARTS   AGE
pod/ad-algorithm-selector-d4f776d88-8m2h2                 1/1     Running     0          9h
pod/ade-infra-clients-7df779d9b8-pkvjc                    1/1     Running     0          39h
pod/ade-service-id-job-lm7f6                              0/1     Completed   0          8h
pod/ade-tenant-active-job-jmmhw                           0/1     Completed   0          40h
pod/ade-tenant-id-job-zsbqs                               0/1     Completed   0          8h
pod/adeauthsvc-966d56977-6scdh                            1/1     Running     0          40h
pod/adeops-rsso-vzj9d                                     0/1     Completed   0          40h
pod/adereporting-785fc79648-6vpk8                         2/2     Running     0          40h
pod/adereporting-apiservice-6c546555fc-wj6hm              1/1     Running     0          40h
pod/adereporting-content-mm66n                            0/1     Completed   0          40h
pod/adereporting-content-update-82sxl                     0/1     Completed   0          40h
pod/adereporting-job-scheduler-28103400-h2nxb             0/1     Completed   0          12m
```

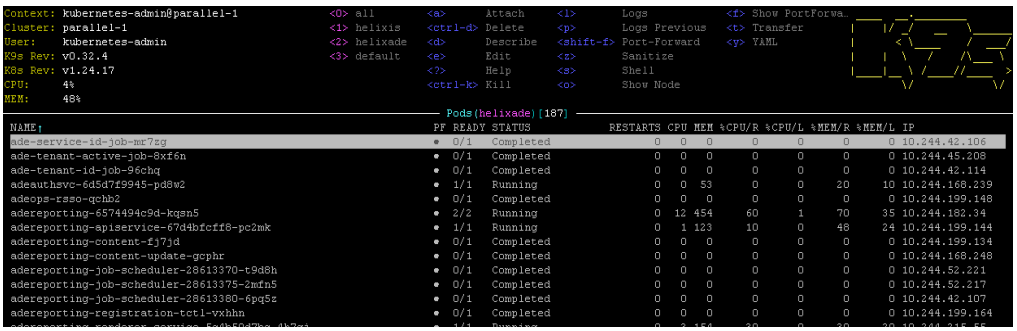
A statefulset with 1 of 2 ready could indicate a problem, but in this case, it is only 2 minutes and 44 seconds old, so it is likely still provisioning. No cause for alarm, but worth watching by running the following command:

```
kubectl -n helixade get po -w
```

- You can also "tail" the end of the deployment.log in the logs folder (/root/ITOM/helix-on-prem-deployment-manager/logs) to get more detailed info about how the installer is going. This is also where any errors will show up.

- K9s is also useful for watching pods or quickly checking container logs. To open view pods in the helixade namespace, type the following and hit the Enter button on your keyboard (lowercase)

```
k9s -n helixade
```




3. A successful deployment will show the message "Completed Helix On-prem Installation" in the installer.

```
!!!!!!!!!!!!!!!!!!!!!!!!          Completed Helix On-prem Installation          !!!!!!!!!!!!!!!!!!!!!!!!
[root@deploymentengine helix-on-prem-deployment-manager]#
```

---

*TROUBLESHOOTING*

---

**Refer to the following Communities Blog for examples how to troubleshoot pods in Kubernetes**:

https://community.bmc.com/s/news/aA33n000000CvAaCAK/troubleshooting-applications-on-kubernetes

**Troubleshooting all pods in each app deploy:**

It's common to need to pull logs from all containers within a given app label. To see labels on a single pod, use "kubectl describe" with the pod name (note the suffix in the pod name will be different in each install so a 'kubectl get pods -n <namespace> | grep <prefix>' will help find the name you need).

```
[root@control1 helix-on-prem-deployment-manager]# kubectl -n helixade describe pod rsso-7b97cb9b84-4bktp
```

**Uninstalling:**

If the installation process fails or gets interrupted, the approach to resolve it depends on the stage where the failure occurred. Logs are essential for investigating the cause, but here's a simplified guide to doing an uninstall if necessary:

1. Uninstall the existing deployment by using the **./uninstall.sh** script located in the same directory as deployment-manager.sh (i.e., the helix-on-prem-deployment-manager directory). In this training environment, it will take ~10 minutes.
2. Verify if any pods are still running: **Kubectl get pods -n helixade** (wait until this command returns nothing)
3. Re-deploy the application from the beginning.

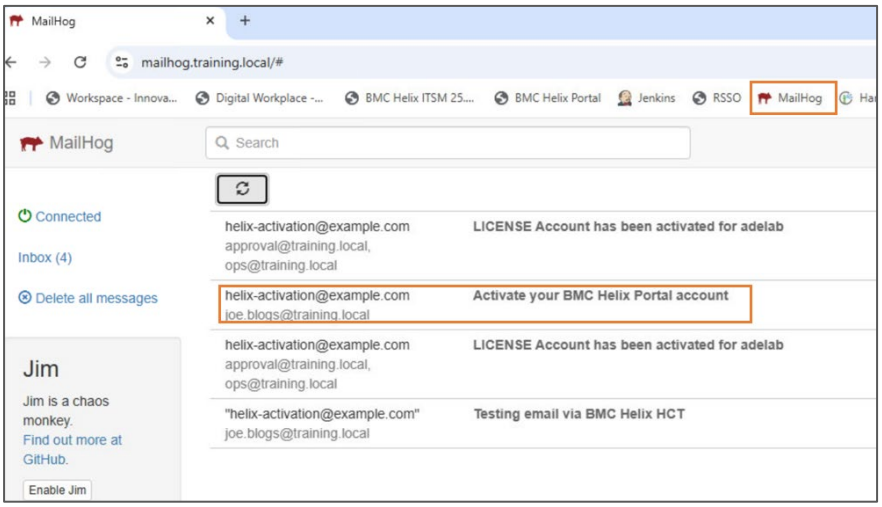
## Task 9: Use Mailhog Email to Activate BMC Helix Portal account

After the BMC Helix Platform Common Service are deployed, the tenant administrator receives the following emails:

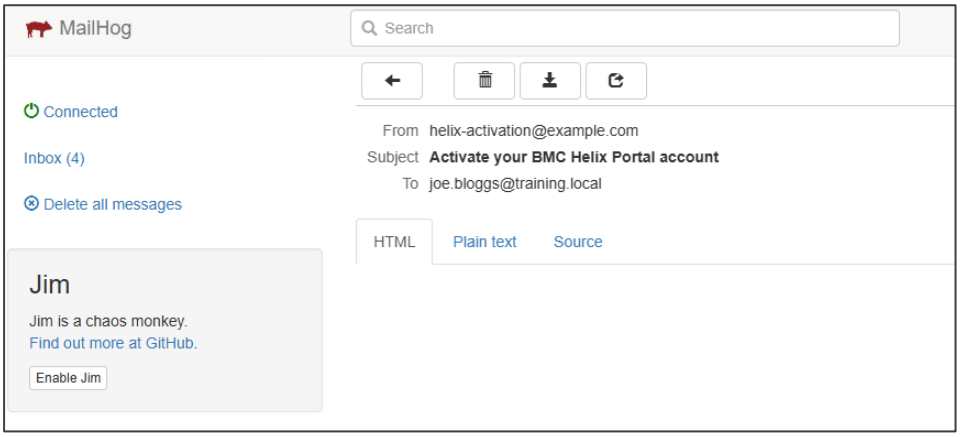
- An email with details about the BMC Helix Platform account
- An email to change the BMC Helix Platform account password at the first login

Use Mailhog (an email testing tool that captures outgoing messages from applications and displays them in a web interface) to access the email that will allow you to activate the hannah_admin account and set the password. Finally, log into the Helix Portal to confirm access.

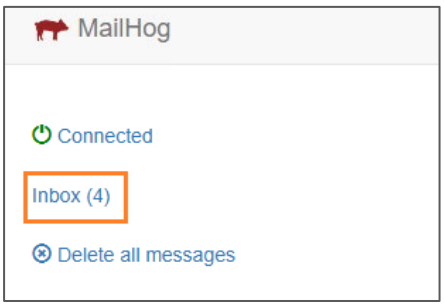
1. On your landing console, open the **Chrome** browser and click the **MailHog** shortcut.




2. Open the **Activate your BMC Helix Portal account** message. In our training environment the HTML tab may first appear blank.

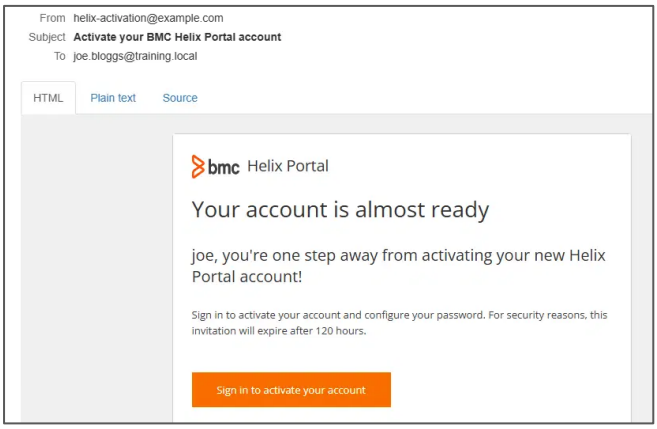



3. If the HTML tab is blank, click the **Inbox (4)** link to refresh the inbox. Once the browser page refreshes, reopen the **Activate your BMC Helix Portal account** message and view the content on the HTML tab.

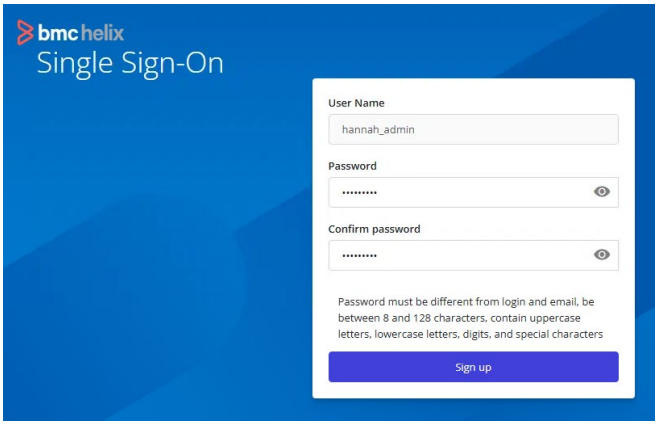



**Note:** This step is necessary for Chrome to load the required HTML content in the lab environment to complete the remaining lab steps.

4. Click on the orange **Sign in to activate your account** button, which will open a new tab in the browser window.

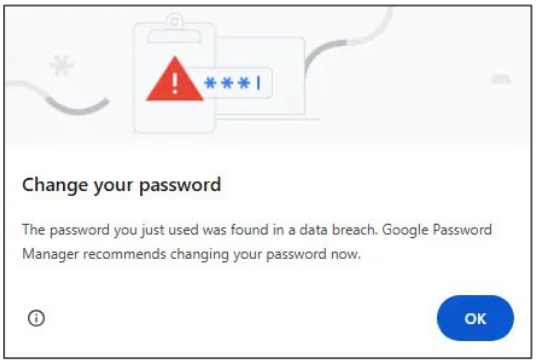



5. Set **hannah_admin** password to **Passw0rd!**
   **Note:** Be careful not to copy/paste a white space as we will be reusing this value later.

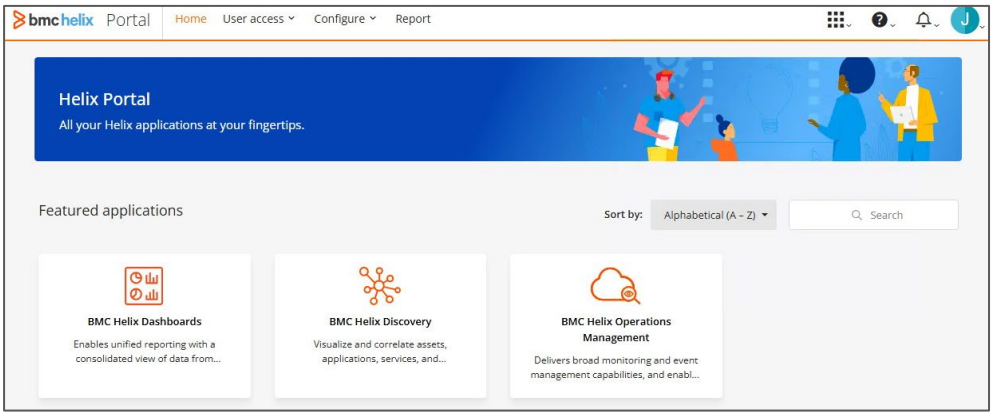



6. Click **Sign up.**

7. Click **OK** to dismiss the Chrome warning to change your password.




8. Using the credentials you just set for **hannah_admin (hannah_admin/Passw0rd!)**, login to the Helix Portal, and verify the Dashboards, Discovery and Helix Operations Management tiles display.


bmc helix Portal
Home
User access
Configure
Report
Helix Portal
All your Helix applications at your fingertips.
Featured applications
Sort by:
Alphabetical (A – Z)
Search
BMC Helix Dashboards
Enables unified reporting with a consolidated view of data from...
BMC Helix Discovery
Visualize and correlate assets, applications, services, and...
BMC Helix Operations Management
Delivers broad monitoring and event management capabilities, and enabl...